

Notes on Gauge Theory

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Preface

本次讨论班的内容主要是规范理论，我们将会介绍规范理论所需的几何基础、用几何语言描述规范理论，以及用规范理论研究低维拓扑。

首先我们会介绍规范理论的几何基础，如联络、主丛、特征类等，这些内容在物理学中是非常重要的工具，在数学中也有着广泛的应用。我们将从黎曼几何中的 Levi-Civita 联络开始，逐渐引入更一般的仿射联络和主丛上的联络，并介绍它们的曲率和特征类等相关内容。

物理学家用联络描述规范场论中的规范场，曲率则对应场强，我们将会用几何语言描述电磁理论和 Yang-Mills 理论等规范场论的基本内容。

最后，数学家用规范理论构造了很多重要的拓扑不变量，如 Donaldson 不变量、Seiberg-Witten 不变量等，我们将会介绍这些不变量的定义，以及它们在低维拓扑中的应用。

以下是本次讨论班的参考资料：

- 《Differential Geometry - Connections, Curvature and Characteristic Classes》(Loring W. Tu)
- 《黎曼几何引论 (上下两册)》(陈维桓, 李兴校)
- 《Lectures on Seiberg-Witten Invariants》(John Douglas Moore)
- 何思奇老师在陈所暑期学校的讲义 <http://www.cim.nankai.edu.cn/2020/0709/c11453a284191/page.htm>

0 Prologue: Electromagnetism and $U(1)$ Gauge Theory

0.1 Maxwell's equations

We begin the story with Maxwell's equations. Recall that Maxwell's equations in vacuum are

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases}$$

Here

$$\begin{aligned} \mathbf{E} &= (E_x, E_y, E_z) \quad \text{is the electric field,} \\ \mathbf{B} &= (B_x, B_y, B_z) \quad \text{is the magnetic field.} \end{aligned}$$

Each component is a function of the spacetime coordinates (t, x, y, z) .

Below we will list some concepts and notations commonly used in physics:

- 4-vector potential A_μ ¹

$$A_\mu = (\phi, A_x, A_y, A_z),$$

where ϕ is the *scalar potential* and $\mathbf{A} = (A_x, A_y, A_z)$ is the *vector potential*.

- Electromagnetic field strength tensor $F_{\mu\nu}$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

- Relation between A_μ and $F_{\mu\nu}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

¹ $\mu = 0, 1, 2, 3$ are the spacetime indices, where 0 corresponds to the time coordinate t , and 1, 2, 3 correspond to the spatial coordinates x, y, z .

²Here we use the notation $\partial_0 := \frac{\partial}{\partial t}, \partial_1 := \frac{\partial}{\partial x}, \partial_2 := \frac{\partial}{\partial y}, \partial_3 := \frac{\partial}{\partial z}$.

- *Gauge transformation of the 4-vector potential*

$$A_\mu \mapsto A_\mu + \partial_\mu \chi \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \text{grad } \chi. \end{cases}$$

where χ is an arbitrary smooth function. One can check that this transformation leaves the electromagnetic field strength tensor $F_{\mu\nu}$ invariant.

Although Maxwell's equations are already quite concise, we can still transform them into an even more elegant expression using the language of differential forms.

Let $\mathbb{R}^{3,1}$ be the spacetime with coordinates (t, x, y, z) , equipped with the Minkowski metric

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2.$$

For simplicity, we set the speed of light $c = 1$.

The Hodge star operator $*$ on $\Omega^*(\mathbb{R}^{3,1})$ is defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle dt \wedge dx \wedge dy \wedge dz, \quad \forall \alpha, \beta \in \Omega^*(\mathbb{R}^{3,1}).$$

Explicitly, we have

$$\begin{aligned} *1 &= dt \wedge dx \wedge dy \wedge dz, \\ *dt &= dx \wedge dy \wedge dz, \\ *dx &= dt \wedge dy \wedge dz, \\ *dy &= dt \wedge dz \wedge dx, \\ *dz &= dt \wedge dx \wedge dy, \\ *(dt \wedge dx) &= -dy \wedge dz, \\ *(dt \wedge dy) &= -dz \wedge dx, \\ *(dt \wedge dz) &= -dx \wedge dy, \\ *(dx \wedge dy) &= dt \wedge dz, \\ *(dy \wedge dz) &= dt \wedge dx, \\ *(dz \wedge dx) &= dt \wedge dy, \\ *(dt \wedge dx \wedge dy) &= dz, \\ *(dt \wedge dy \wedge dz) &= dx, \\ *(dt \wedge dz \wedge dx) &= dy, \\ *(dx \wedge dy \wedge dz) &= dt, \\ *(dt \wedge dx \wedge dy \wedge dz) &= -1. \end{aligned}$$

It is direct to check that

$$*^2 = (-1)^{p+1} \quad \text{on } \Omega^p(\mathbb{R}^{3,1}).$$

Now we can introduce the electromagnetic field strength 2-form

$$F = -dt \wedge (E_x dx + E_y dy + E_z dz) + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$$

Then

$$\begin{aligned} dF = & dt \wedge \left[(\partial_x E_y - \partial_y E_x) dx \wedge dy + (\partial_y E_z - \partial_z E_y) dy \wedge dz + (\partial_z E_x - \partial_x E_z) dz \wedge dx \right] \\ & + dt \wedge \left(\partial_t B_x dy \wedge dz + \partial_t B_y dz \wedge dx + \partial_t B_z dx \wedge dy \right) \\ & + \left(\partial_x B_x + \partial_y B_y + \partial_z B_z \right) dx \wedge dy \wedge dz, \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \end{cases} \iff dF = 0.$$

On the other hand,

$$*F = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy + dt \wedge (B_x dx + B_y dy + B_z dz).$$

So

$$\begin{aligned} d * F = & -dt \wedge \left[(\partial_x B_y - \partial_y B_x) dx \wedge dy + (\partial_z B_x - \partial_x B_z) dz \wedge dx + (\partial_y B_z - \partial_z B_y) dy \wedge dz \right] \\ & + dt \wedge \left(\partial_t E_x dy \wedge dz + \partial_t E_y dz \wedge dx + \partial_t E_z dx \wedge dy \right) \\ & + \left(\partial_x E_x + \partial_y E_y + \partial_z E_z \right) dx \wedge dy \wedge dz. \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff d * F = 0.$$

Since $* : \Omega^p(\mathbb{R}^{3,1}) \rightarrow \Omega^{4-p}(\mathbb{R}^{3,1})$ is an isomorphism,

$$d * F = 0 \iff *d * F = 0.$$

Therefore,

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \begin{cases} dF = 0, \\ d * F = 0. \end{cases}$$

Here $d^* = *d*$ is the dual of d .

In Riemannian geometry, a p -form α is called *harmonic* if $d\alpha = 0$ and $d^*\alpha = 0$. Thus Maxwell's equations in vacuum corresponds to a harmonic 2-form F .

Poincaré lemma says all closed forms on $\mathbb{R}^{3,1}$ are exact. So if $dF = 0$, there exists a 1-form A such that

$$F = dA.$$

If we write

$$A = \phi dt + A_x dx + A_y dy + A_z dz,$$

Then

$$F = dA \implies \begin{cases} E_x = \partial_x \phi - \partial_t A_x, \\ E_y = \partial_y \phi - \partial_t A_y, \\ E_z = \partial_z \phi - \partial_t A_z, \\ B_x = \partial_y A_z - \partial_z A_y, \\ B_y = \partial_z A_x - \partial_x A_z, \\ B_z = \partial_x A_y - \partial_y A_x. \end{cases} \implies \begin{cases} \mathbf{E} = \text{grad } \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \text{curl } \mathbf{A}. \end{cases}$$

It is easy to see that A is not unique. In fact, if we transform $A \mapsto A + d\chi$ for some smooth function χ , then

$$F \mapsto dA + d^2\chi = dA = F.$$

Explicitly,

$$A \mapsto A + d\chi \implies \begin{cases} \phi \mapsto \phi + \partial_t \chi, \\ A_x \mapsto A_x + \partial_x \chi, \\ A_y \mapsto A_y + \partial_y \chi, \\ A_z \mapsto A_z + \partial_z \chi. \end{cases} \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \nabla \chi. \end{cases}$$

We can summarize the above discussion in a table:

Physics	Differential Geometry
4-vector potential A_μ	1-form A
Electromagnetic field strength tensor $F_{\mu\nu}$	2-form F
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F = dA$
Gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$	$A \mapsto A + d\chi$
Maxwell's equations	$dF = 0, d^*F = 0$

0.2 $U(1)$ gauge theory

Let $\psi(x)$ be a complex-valued function on $\mathbb{R}^{3,1}$, which represents the wave function of a matter field. Here x denotes the spacetime coordinates. In physics, $|\psi(x)|^2$ represents the probability amplitude of finding the particle at a given point in spacetime. If we transform

$$\psi(x) \mapsto \psi'(x) = e^{i\chi} \psi(x)$$

for some real constant χ . This transformation is called a *global $U(1)$ gauge transformation*. Since the probability amplitude $|\psi'(x)|^2 = |\psi(x)|^2$ remains the same, the physics is invariant under global $U(1)$ gauge transformations.

Now we want to promote the global $U(1)$ gauge symmetry to a local one. That is, we want the physics to be invariant under the transformation

$$\psi(x) \mapsto \psi'(x) = e^{i\chi(x)} \psi(x)$$

for some smooth function $\chi \in C^\infty(\mathbb{R}^{3,1})$.

For some physical reason, we want the derivative of ψ to be covariant under transformations. This holds in the case of global $U(1)$ gauge transformations, since

$$d\psi \mapsto d\psi' = e^{i\chi} d\psi.$$

However, in the case of local $U(1)$ gauge transformations,

$$d\psi \mapsto d\psi' = e^{i\chi(x)} d\psi + (de^{i\chi(x)})\psi.$$

The extra term $(de^{i\chi(x)})\psi$ breaks the covariance of the derivative.

To fix this problem, we introduce a 1-form A and define the *covariant derivative* by

$$\nabla\psi = d\psi + iA\psi.$$

Under the local $U(1)$ gauge transformation,

$$\nabla \mapsto \nabla' = d + iA',$$

we want $\nabla\psi$ to be covariant, i.e.

$$\nabla\psi \mapsto \nabla'\psi' = e^{i\chi(x)} \nabla\psi.$$

Then by

$$\begin{aligned} \nabla'\psi' &= d\psi' + iA'\psi' = e^{i\chi(x)} d\psi + ie^{i\chi(x)}(A' + d\chi)\psi, \\ e^{i\chi(x)} \nabla\psi &= e^{i\chi(x)}(d\psi + iA\psi) = e^{i\chi(x)} d\psi + ie^{i\chi(x)} A\psi, \end{aligned}$$

we must have

$$A' = A - d\chi.$$

Ignoring the sign, this is exactly the gauge transformation of the electromagnetic potential A in electromagnetism. Thus, the electromagnetic potential A can be interpreted as a gauge field that ensures the covariance of the derivative under local $U(1)$ gauge transformations.

1 Affine Connections

1.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

1.1.1 Covariant Derivative

Definition 1.1. A connection ∇ on a manifold M is a map

$$\begin{aligned}\nabla : C^\infty(M, TM) \times C^\infty(M, TM) &\rightarrow C^\infty(M, TM) \\ (Y, X) &\mapsto \nabla_Y X\end{aligned}$$

satisfying the following properties:

$$\begin{aligned}\nabla_{f_1 Y_1 + f_2 Y_2} X &= f_1 \nabla_{Y_1} X + f_2 \nabla_{Y_2} X, \\ \nabla_Y (\lambda_1 X_1 + \lambda_2 X_2) &= \lambda_1 \nabla_Y X_1 + \lambda_2 \nabla_Y X_2, \\ \nabla_Y f X &= df(Y) \cdot X + f \nabla_Y X.\end{aligned}$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 1.2. The **torsion**¹ of a connection ∇ is defined as the tensor

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y].$$

The **curvature** of a connection ∇ is defined as the tensor

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Definition 1.3. A connection ∇ of a Riemannian manifold (M, g) is called a Levi-Civita connection if it is torsion-free and compatible with the metric g , i.e.

$$\begin{aligned}\nabla_X Y - \nabla_Y X &= [X, Y], \\ X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z).\end{aligned}$$

¹It can only be defined for connections on the tangent bundle, since it involves both $\nabla_X Y$ and $\nabla_Y X$.

1.1.2 Exterior covariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned}\nabla : C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M \otimes TM) \\ X &\mapsto \nabla X\end{aligned}$$

where ∇X is a 1-form taken value in TM^1 and $(\nabla X)(Y) := \nabla_Y X$. ∇ is not a $C^\infty(M)$ -linear map, but

$$\nabla(fX) = df \otimes X + f\nabla X.$$

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, TM) &\rightarrow \Omega^{*+1}(M, TM)^2 \\ \eta &\mapsto \nabla \eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, TM),$$

For example, let $\eta = \alpha \otimes X$ be a section of $\Omega^r(M, TM)$, then

$$\nabla \eta = \nabla(\alpha \otimes X) = d\alpha \otimes X + (-1)^r \alpha \wedge \nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla \circ \nabla, \nabla^3, \dots, \nabla^k$, and we can prove the following property:

Proposition 1.4. $\nabla^2 : \Omega^*(M, TM) \rightarrow \Omega^{*+2}(M, TM)$ is an $\Omega^*(M)$ -linear map, i.e. for any $\alpha \in \Omega^*(M), \eta \in \Omega^*(M, TM)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta.$$

Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \text{End}(TM))$, which is called the **curvature** of the connection ∇ .

Proof.

$$\begin{aligned}\nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge d^\nabla \eta) \\ &= d^2 \alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge d^\nabla \eta + (-1)^q d\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ &= \alpha \wedge \nabla^2 \eta.\end{aligned} \quad \square$$

Unlike the usual exterior derivative d , we have $\nabla^2 \neq 0$ in general. The following proposition explains why ∇^2 is called the curvature of ∇ :

Proposition 1.5. For $X, Y, Z \in C^\infty(M, TM)$, we have

$$(\nabla^2 Z)(X, Y) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Proof. We will prove it after introducing the connection 1-forms. $\square$

¹One can simply view ∇X as a $(1, 1)$ -tensor field.

² $\Omega^r(M, TM) := C^\infty(M, \wedge^r T^*M \otimes TM)$.

1.1.3 Connection 1-form

Locally, we always have a local frame. Let $\{e_i\}$ be a local frame of TM on an open set $U \subset M$, and $\{\theta^i\}$ be the dual frame. then we can write

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k,$$

where Γ_{ij}^k are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega_j^k := \Gamma_{ij}^k \theta^i,$$

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)^1$, then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_n) = (e_1, \dots, e_n) \omega = (e_1, \dots, e_n) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \cdots & \omega_n^n \end{pmatrix}^2$$

We can compute the curvature in terms of the connection 1-form ω :

$$\begin{aligned} \nabla^2(e_1, \dots, e_n) &= \nabla(\nabla(e_1, \dots, e_n)) = \nabla((e_1, \dots, e_n) \omega) \\ &= (\nabla(e_1, \dots, e_n)) \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) \omega \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) (d\omega + \omega \wedge \omega). \end{aligned}$$

Define

$$\Omega := d\omega + \omega \wedge \omega, \tag{1}$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 1.6. *Let $R(e_i, e_j)e_k = R_{ijk}^l e_l$, then*

$$\Omega_k^l(e_i, e_j) = R_{ijk}^l.$$

¹The upper index k is the row index, and the lower index j is the column index.

²This is an awkward notation. Due to certain notational reasons, we consider arranging the frame as row vectors, which results in differential forms acting on the right of vectors. For ordinary vectors, we are free to place differential forms on the left or right without introducing sign changes. However, it should be noted that for vector-valued differential forms, placing the differential form on the left or right does make a sign difference.

Proof. Since $\Omega_k^l = d\omega_k^l + \omega_m^l \wedge \omega_k^m$,

$$\begin{aligned} (d\omega_k^l)(e_i, e_j)e_l &= e_i(\omega_k^l(e_j))e_l - e_j(\omega_k^l(e_i))e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}(\omega_k^l(e_j)e_l) - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega_k^l(e_i)e_l) + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}\nabla_{e_j}e_k - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l, \end{aligned}$$

and

$$\begin{aligned} (\omega_m^l \wedge \omega_k^m)(e_i, e_j)e_l &= \omega_m^l(e_i)\omega_k^m(e_j)e_l - \omega_m^l(e_j)\omega_k^m(e_i)e_l \\ &= \omega_k^m(e_j)\nabla_{e_i}e_m - \omega_k^m(e_i)\nabla_{e_j}e_m. \end{aligned}$$

Thus

$$\Omega_k^l(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i, e_j]}e_k = R(e_i, e_j)e_k. \quad \square$$

Corollary 1.7. *This proposition immediately implies Prop 1.5 since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega_k^l(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

The connection 1-form ω is not a tensor, since it depends on the choice of the local frame. Let $\{\tilde{e}_j\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_j\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n) a,$$

then

$$\begin{aligned} (\tilde{e}_1, \dots, \tilde{e}_n)\tilde{\omega} &= \nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla((e_1, \dots, e_n) a) \\ &= (\nabla(e_1, \dots, e_n))a + (e_1, \dots, e_n)da \\ &= (e_1, \dots, e_n)\omega a + (e_1, \dots, e_n)da \\ &= (\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + da). \end{aligned}$$

Thus

$$\tilde{\omega} = a^{-1}da + a^{-1}\omega a. \quad (2)$$

Applying the transformation formula (2) to (1), we get

$$\begin{aligned} \tilde{\Omega} &= d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega} \\ &= d(a^{-1}da + a^{-1}\omega a) + (a^{-1}da + a^{-1}\omega a) \wedge (a^{-1}da + a^{-1}\omega a) \\ &= da^{-1} \wedge da + da^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a, \end{aligned}$$

since

$$a^{-1}a = I \implies da^{-1}a + a^{-1}da = 0 \implies da^{-1} = -a^{-1}da a^{-1},$$

we have

$$\begin{aligned}\tilde{\Omega} &= -a^{-1}da a^{-1} \wedge da - a^{-1}da a^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a \\ &= a^{-1}(d\omega + \omega \wedge \omega)a = a^{-1}\Omega a.\end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \quad (3)$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

For a Riemannian manifold (M, g) , let ∇ be the Levi-Civita connection. We can take $\{e_i\}$ to be a local orthonormal frame, and $\{\omega^i\}$ be the corresponding dual frame. Then

$$\begin{aligned}\langle \nabla_{e_k} e_i, e_j \rangle + \langle e_i, \nabla_{e_k} e_j \rangle &= e_k \langle e_i, e_j \rangle = 0 \\ \implies \langle \omega_i^l(e_k) e_l, e_j \rangle + \langle e_i, \omega_j^l(e_k) e_l \rangle &= 0 \\ \implies \omega_i^j(e_k) + \omega_j^i(e_k) &= 0.\end{aligned}$$

So the connection 1-form ω w.r.t. an orthonormal frame is skew-symmetric, i.e. $\omega \in \Omega^1(M, \mathfrak{so}(n))$. Since $\Omega = d\omega + \omega \wedge \omega$, the curvature 2-form Ω is also skew-symmetric, i.e. $\Omega \in \Omega^2(M, \mathfrak{so}(n))$.

1.2 General Affine Connections

Let $\pi : E \rightarrow M$ be a rank m vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

1.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 1.8. An (affine) connection ∇ on a vector bundle E is a map

$$\nabla : C^\infty(M, TM) \times C^\infty(M, E) \rightarrow C^\infty(M, E)$$

$$(X, s) \mapsto \nabla_X s$$

satisfying

$$\begin{aligned}\nabla_{f_1 X_1 + f_2 X_2} s &= f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s, \\ \nabla_X (\lambda_1 s_1 + \lambda_2 s_2) &= \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2, \\ \nabla_X (fs) &= df(X) \cdot s + f \nabla_X s,\end{aligned}$$

The existence of connections on any vector bundle can be proved by using partitions of unity, which we left as an exercise.

The reason why we call ∇ an **affine** connection is that given two connections ∇_1 and ∇_2 on E , $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, E) &\rightarrow \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla \eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, E).$$

Proposition 1.9. $\nabla^2 : \Omega^*(M, E) \rightarrow \Omega^{*+2}(M, E)$ is an $\Omega^*(M)$ -linear map, i.e.

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta, \quad \forall \alpha \in \Omega^*(M), \eta \in \Omega^*(M, E).$$

Proof. For $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\begin{aligned}\nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta) \\ &= d^2 \alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge \nabla \eta + (-1)^q d\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ &= \alpha \wedge \nabla^2 \eta.\end{aligned} \quad \square$$

Definition 1.10. The curvature of a connection ∇ on E is defined as $R := \nabla^2 \in \Omega^2(M, \text{End}(E))$.

1.2.2 Connection 1-form

Let $\{e_i\}$ be a local frame of E on an open set $U \subset M$. Define ω_j^k by

$$\nabla_X e_j = \omega_j^k(X) e_k,$$

this 1-form matrix $\omega = (\omega_j^k)$ is called **the connection 1-form** of ∇ w.r.t. $\{e_i\}$. Then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_m) = (e_1, \dots, e_m) \omega = (e_1, \dots, e_m) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^m \\ \vdots & & \vdots \\ \omega_m^1 & \cdots & \omega_m^m \end{pmatrix}.$$

1.2.3 Horizontal Distribution

In this subsection, we give another interpretation of a connection on a vector bundle, namely a horizontal distribution, and prove that it is equivalent to the definition given in terms of covariant derivatives.

Let $\pi : E \rightarrow M$ be a rank m vector bundle, and $p \in E_x$ be a point in the fibre over $x \in M$. The projection π induces a linear map $\pi_{*p} : T_p E \rightarrow T_x M$. We define $\mathcal{V}_p := \ker \pi_{*p}$, which is called the **vertical tangent space** at p . Thus we have a short exact sequence

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_p E \xrightarrow{\pi_*} T_{\pi(p)} M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_p E - \dim T_{\pi(p)} M = m.$$

The kernel of π_* form a subbundle of TE . We call it the **vertical subbundle** of TE and denote it by $\mathcal{V}$.

Proposition 1.11. *For each $p \in E$, there is an isomorphism $E_{\pi(p)} \cong \mathcal{V}_p$, which induces an isomorphism $\pi^* E \cong \mathcal{V}$.*

Proof. A point $(p, p') \in \pi^* E$ must satisfy $\pi(p) = \pi(p')$, so we can define a curve $\gamma(t) = p + tp'$ in E . Then $\gamma'(0) \in T_p E$ gives a tangent vector at p , this gives an injective morphism

$$\pi^* E \hookrightarrow TE, \quad (p, p') \mapsto (p, \gamma'(0)).$$

So we can view $\pi^* E$ as a subbundle of TE . Moreover, since $\gamma(t) \subset E_{\pi(p)}$, we have $\pi(\gamma(t)) = \pi(p)$, thus $\pi_{*p}(\gamma'(0)) = 0$. Therefore, $\pi^* E \subset \ker \pi_*$.

Since π_{*p} is a surjective linear map, we have

$$\dim \ker \pi_{*p} = \dim T_p E - \dim T_{\pi(p)} M = m = (\pi^* E)_p.$$

Thus $\pi^* E = \ker \pi_*$. □

For any $p \in E$, we can define a map

$$\iota_p : E_{\pi(p)} \rightarrow \mathcal{V}_p \subset T_p E, \quad p' \mapsto \gamma'(0),$$

where $\gamma(t) = p + tp'$ is a curve in E . Then ι_p is an isomorphism between $E_{\pi(p)}$ and $\mathcal{V}_p$.

Let $\pi^* TM$ be the pull-back of TM by π , then $\pi_* : TE \rightarrow TM$ can be viewed as a morphism $\tilde{\pi}_* : TE \rightarrow \pi^* TM$. The above proposition implies the following short exact sequence of vector bundles over E :

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \xrightarrow{\tilde{\pi}_*} \pi^* TM \longrightarrow 0$$

In smooth category, all short exact sequences of vector bundles split, so we can find a subbundle $\mathcal{H} \hookrightarrow TE$ such that

$$TE = \mathcal{H} \oplus \mathcal{V}, \quad \tilde{\pi}_* : \mathcal{H} \rightarrow \pi^*TM \text{ is an isomorphism.}$$

On the vector bundle E , we have two natural operations: scalar multiplication and addition.

- The scalar multiplication:

$$m : \mathbb{R} \times E \rightarrow E, \quad m(\lambda, p) = \lambda p.$$

When fixing λ , we get a map $m_\lambda : E \rightarrow E$, $m_\lambda(p) = \lambda p$, which is the scalar multiplication by λ .

- The addition:

$$a : E \times_M E \rightarrow E, \quad a(p, q) = p + q,$$

where $E \times_M E = \{(p, q) \in E \times E : \pi(p) = \pi(q)\}$.

Definition 1.12. A **horizontal distribution** of E is a smooth subbundle $\mathcal{H} \subset TE$ satisfying the following properties

1. $\tilde{\pi}_* : \mathcal{H} \rightarrow \pi^*TM$ is an isomorphism, or equivalently, there exists a splitting map $k : \pi^*TM \rightarrow TE$ such that $\tilde{\pi}_* \circ k = \text{id}$.

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \begin{array}{c} \xrightarrow{\tilde{\pi}_*} \\ \xleftarrow{k} \end{array} \pi^*TM \longrightarrow 0$$

2. $\mathcal{H}$ is invariant under the scalar multiplication on E , i.e. for any $\lambda \in \mathbb{R}$, $p \in E$,

$$m_{\lambda * p} \mathcal{H}_p = \mathcal{H}_{\lambda p}.$$

3. $\mathcal{H}$ is invariant under the addition on E , i.e. for any $(p, q) \in E \times_M E$,

$$a_{*(p,q)}(\mathcal{H}_p, \mathcal{H}_q) = \mathcal{H}_{p+q}.$$

The last two conditions are saying that $\mathcal{H}$ is compatible with the linear structure of E .

There is no canonical way to choose a horizontal distribution, so a horizontal distribution is an extra structure on E .

In what follows, we reveal that covariant derivatives and horizontal distributions can actually determine each other.

1.2.4 Covariant Derivative determines Horizontal Distribution

Let ∇ be a connection on E . Given any smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = x$, we say $\tilde{\gamma} : (-\varepsilon, \varepsilon) \rightarrow E$ is a **lift** of γ if $\pi(\tilde{\gamma}(t)) = \gamma(t)$ for all t .

Definition 1.13. A lift $\tilde{\gamma}$ of γ is called a **horizontal lift** if

$$\nabla_{\gamma'(t)} \tilde{\gamma}(t) = 0, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Example 1.14. In the case of a Riemannian manifold (M, g) , the horizontal lift is exactly the parallel transport.

Proposition 1.15. For any (piecewise) smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ and any point $p \in E_{\gamma(0)}$, there exists a unique horizontal lift $\tilde{\gamma}$ of γ starting at p .

Sketch of proof. The proof is exactly the same as the existence and uniqueness of geodesics. The outline of the proof is to write the equations as a system of ODEs in a local frame; the conclusion then follows from the existence and uniqueness theory of linear ODEs. $\square$

For any vector $u \in T_x M$ and any point $p \in E_x$, we can find a curve γ such that $\gamma(0) = x$ and $\gamma'(0) = u$, the horizontal lift $\tilde{\gamma}$ gives a tangent vector $\tilde{\gamma}'(0) \in T_p E$.

Fact 1.1. $\tilde{\gamma}'(0) \in T_p E$ is independent of the choice of γ .

Sketch of proof. By solving the ODEs, we know that $\tilde{\gamma}'(0)$ depends only on the derivative of γ at 0, which is $\gamma'(0) = u$. $\square$

So there is a well-defined map

$$T_{\pi(p)} M \rightarrow T_p E, \quad (u^h)_p := \tilde{\gamma}'(0).$$

We call $(u^h)_p$ the **horizontal lift** of u at p . By definition,

$$\pi(\tilde{\gamma}(t)) = \gamma(t) \implies \pi_{*p}(u^h)_p = \pi_{*p}(\tilde{\gamma}'(0)) = \gamma'(0) = u.$$

So we obtain a linear injection from $T_{\pi(p)} M$ to $T_p E$, whose image forms a distribution on E denoted by $\mathcal{H}$, i.e.,

$$\mathcal{H} := \{(p, (u^h)_p) \mid p \in E, u \in T_{\pi(p)} M\}.$$

Fact 1.2. $\mathcal{H}$ is a smooth subbundle of TE .

Sketch of proof. This fact relies on the smooth dependence of solutions of linear ODEs on initial conditions and parameters. $\square$

Fact 1.3. $\mathcal{H}$ is invariant under the scalar multiplication and addition on E .

Proof. Let $p, q \in E$ such that $\pi(p) = \pi(q)$, $\lambda \in \mathbb{R}$, and γ be a curve such that $\gamma(0) = \pi(p)$. Let $\tilde{\gamma}_p, \tilde{\gamma}_q$ be the horizontal lift of γ starting at p, q respectively, then by the uniqueness of horizontal lift, $\lambda\tilde{\gamma}_p$ is the horizontal lift of γ starting at λp , and $\tilde{\gamma}_p + \tilde{\gamma}_q$ is the horizontal lift of γ starting at $p + q$. Thus

$$\begin{aligned} m_{\lambda * p} \tilde{\gamma}'(0) &= (\lambda \tilde{\gamma}_p)'(0) \in \mathcal{H}_{\lambda p}, \\ a_{*(p,q)}(\tilde{\gamma}'_p(0), \tilde{\gamma}'_q(0)) &= (\tilde{\gamma}_p + \tilde{\gamma}_q)'(0) \in \mathcal{H}_{p+q}. \end{aligned} \quad \square$$

The above discussion is summarized in the following theorem.

Theorem 1.16. $\mathcal{H}$ is a horizontal distribution of E .

Example 1.17. For a Riemannian manifold (M, g) and the Levi-Civita connection ∇ , the tangent bundle $T(TM)$ has a decomposition $T(TM) = \mathcal{V} \oplus \mathcal{H}$. We can define a metric G on TM by

$$\begin{aligned} G(X^h, Y^h) &:= g(\pi_*(X^h), \pi_*(Y^h)), \quad G(X^v, Y^v) := g(\iota^{-1}(X^v), \iota^{-1}(Y^v)), \\ G(X^h, Y^v) &:= 0, \end{aligned}$$

where $X^h, Y^h \in \mathcal{H}$ and $X^v, Y^v \in \mathcal{V}$. G is called **the Sasaki metric** on TM .

1.2.5 Horizontal Distribution determines Covariant Derivative

Conversaly, given a horizontal distribution $\mathcal{H}$, we can derive a connection ∇ on E as follows.

Definition 1.18. For any $p \in E$ and $u \in TE$, we can write $u = h(u) + v(u)$ where $h(u) \in \mathcal{H}$ and $v(u) \in \mathcal{V}$. We call $h(u)$ and $v(u)$ the horizontal and vertical component of u respectively.

For any $X \in C^\infty(M, TM)$ and $s \in C^\infty(M, E)$, the tangent map of s gives a map $s_* : TM \rightarrow TE$, then at any point $x \in M$ we can define $(\nabla_X s)(x)$ by

$$(\nabla_X s)(x) := \iota_{s(x)}^{-1} (v(s_{*x} X_x)).$$

Note that $\iota_{s(x)} : E_x \rightarrow \mathcal{V}_{s(x)}$ is the isomorphism defined above. The idea is taking the vertical component of $s_{*x}(X_x)$.

Theorem 1.19. ∇ defined above is a connection on E .

To prove this theorem, we need the following lemma.

Lemma 1.20. For $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u, w \in E_{\pi(p)}$, we have

$$\begin{aligned} m_{\lambda * p} \iota_p(u) &= \iota_{\lambda p}(\lambda u), \\ a_{*(p,q)}(\iota_p(u), \iota_q(w)) &= \iota_{p+q}(u + w). \end{aligned}$$

Proof. By the definition of ι_p ,

$$\begin{aligned} m_{\lambda * p} \iota_p(u) &= \left. \frac{d}{dt} \right|_{t=0} m_{\lambda}(p + tu) = \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\lambda u) = \iota_{\lambda p}(\lambda u), \\ a_{*(p,q)}(\iota_p(u), \iota_q(w)) &= \left. \frac{d}{dt} \right|_{t=0} (p + tu) + (q + tw) = \iota_{p+q}(u + w). \quad \square \end{aligned}$$

Lemma 1.21. For any $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u \in TE_p$, $w \in TE_q$,

$$\begin{aligned} h(m_{\lambda * p} u) &= m_{\lambda * p}(h(u)), \quad v(m_{\lambda * p} u) = m_{\lambda * p}(v(u)), \\ h(a_{*(p,q)}(u, w)) &= a_{*(p,q)}(h(u), h(w)), \quad v(a_{*(p,q)}(u, w)) = a_{*(p,q)}(v(u), v(w)). \end{aligned}$$

Proof. Decompose u and $m_{\lambda * p} u$ into their horizontal and vertical components, we get

$$m_{\lambda * p}(h(u)) + m_{\lambda * p}(v(u)) = m_{\lambda * p}(u) = h(m_{\lambda * p} u) + v(m_{\lambda * p} u).$$

Since $\mathcal{H}$ is compatible with the linear structure of the vector bundle, we have $m_{\lambda * p}(h(u)) \in \mathcal{H}_{\lambda p}$. Besides, Lemma 1.20 implies $m_{\lambda * p}(v(u)) \in \mathcal{V}_{\lambda p}$, thus

$$m_{\lambda * p}(h(u)) = h(m_{\lambda * p} u), \quad m_{\lambda * p}(v(u)) = v(m_{\lambda * p} u).$$

The proof for the addition is exactly the same. $\square$

Lemma 1.22. For $\nu \in T_{\lambda} \mathbb{R} \cong \mathbb{R}$ and $u \in T_p E$, we have

$$m_{*(\lambda,p)}(\nu, u) = \nu \cdot \iota_p(p) + m_{\lambda * p} u.$$

Proof. let $\gamma(t)$ be a curve in E such that $\gamma(0) = p$ and $\gamma'(0) = u$, then

$$\begin{aligned} m_{*(\lambda,p)}(\nu, u) &= \left. \frac{d}{dt} \right|_{t=0} (\lambda + t\nu)\gamma(t) \\ &= \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\nu p) + \left. \frac{d}{dt} \right|_{t=0} \lambda \gamma(t) \\ &= \nu \cdot \iota_p(p) + m_{\lambda * p} u. \quad \square \end{aligned}$$

Proof of Theorem 1.19. We need to verify that ∇ satisfies the properties of a connection.

- $\nabla_{f_1 X_1 + f_2 X_2} s = f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s$:

$$\begin{aligned}
\nabla_{f_1 X_1 + f_2 X_2} s(x) &= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1 X_1 + f_2 X_2)) \right) \\
&= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1(x) X_{1x})) + v(s_{*x}(f_2(x) X_{2x})) \right) \\
&= \iota_{s(x)}^{-1} \left(f_1(x) v(s_{*x} X_{1x}) + f_2(x) v(s_{*x} X_{2x}) \right) \\
&= f_1 \nabla_{X_1} s(x) + f_2 \nabla_{X_2} s(x).
\end{aligned}$$

- $\nabla_X(\lambda_1 s_1 + \lambda_2 s_2) = \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2$:

For $\mathbb{R}$ -linearity,

$$\begin{aligned}
\nabla_X(\lambda s)(x) &= \iota_{(\lambda s)(x)}^{-1} \left(v((\lambda s)_{*x}(X_x)) \right) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(v(m_{\lambda * s(x)} s_{*x}(X_x)) \right) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(m_{\lambda * s(x)} v(s_{*x}(X_x)) \right) \quad (\text{By Lemma 1.21}) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(\iota_{(\lambda s)(x)}(\lambda \nabla_X s(x)) \right) \quad (\text{By Lemma 1.20}) \\
&= \lambda \nabla_X s(x),
\end{aligned}$$

For additivity,

$$\begin{aligned}
\nabla_X(s_1 + s_2)(x) &= \iota_{(s_1 + s_2)(x)}^{-1} \left(v((s_1 + s_2)_{*x} X_x) \right) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(v(a_{*(s_1(x), s_2(x))}(s_{1*x} X_x, s_{2*x} X_x)) \right) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(a_{*(s_1(x), s_2(x))}(v(s_{1*x} X_x), v(s_{2*x} X_x)) \right) \quad (\text{By Lemma 1.21}) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(\iota_{(s_1 + s_2)(x)}(\nabla_X s_1(x) + \nabla_X s_2(x)) \right) \quad (\text{By Lemma 1.20}) \\
&= \nabla_X s_1(x) + \nabla_X s_2(x).
\end{aligned}$$

- $\nabla_X(fs) = df(X) \cdot s + f \nabla_X s$:

$$\begin{aligned}
\nabla_X(fs)(x) &= \iota_{(fs)(x)}^{-1} \left(v((fs)_{*x} X_x) \right) \\
&= \iota_{(fs)(x)}^{-1} \left(v(m_{f(x)*s(x)}(f_{*x} X_x, s_{*x} X_x)) \right) \\
&= \iota_{(fs)(x)}^{-1} \left(df(X)_x \cdot \iota_{(fs)(x)}(s(x)) + v(m_{f(x)*s(x)} s_{*x} X_x) \right) \quad (\text{By Lemma 1.22}) \\
&= \iota_{(fs)(x)}^{-1} \left(\iota_{(fs)(x)}(df(X)_x \cdot s(x) + f(x) \nabla_X s(x)) \right) \quad (\text{By Lemma 1.20}) \\
&= df(X)_x \cdot s(x) + f(x) \nabla_X s(x). \quad \square
\end{aligned}$$

Exercises

Exercise 1.1. In Riemannian geometry, we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s) -tensor field T , the covariant derivative $\nabla_Z T$ is

$$\begin{aligned} (\nabla_Z T)(X_1, \dots, X_r, \omega^1, \dots, \omega^s) &:= Z(T(X_1, \dots, X_r, \omega^1, \dots, \omega^s)) \\ &\quad - \sum_{i=1}^r T(X_1, \dots, \nabla_Z X_i, \dots, X_r, \omega^1, \dots, \omega^s) \\ &\quad - \sum_{j=1}^s T(X_1, \dots, X_r, \omega^1, \dots, \nabla_Z \omega^j, \dots, \omega^s). \end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X, Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X, Y]} T.$$

We define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s + 1)$ -tensor field

$$(\nabla_{\text{tensor}} T)(Z, \dots, X_i, \dots, \dots, \omega^j, \dots) := (\nabla_Z T)(\dots, X_i, \dots, \dots, \omega^j, \dots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

Prove that for a torsion-free connection, ∇_{tensor} satisfies **the Ricci identity**

$$(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) - (\nabla_{\text{tensor}}^2 T)(Y, X, \dots, \dots) = (R(X, Y)T)(\dots, \dots)$$

Exercise 1.2. For a vector field $Z \in C^\infty(M, TM)$, suppose ∇ is torsion-free. Both $\nabla^2 Z$ and $\nabla_{\text{tensor}}^2 Z$ are $(1, 2)$ -tensor fields, but by Ricci identity, we have

$$(\nabla_{\text{tensor}}^2 Z)(X, Y) - (\nabla_{\text{tensor}}^2 Z)(Y, X) = R(X, Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?

Exercise 1.3 (Torsion 2-form). Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. Define

$$\tau^i := d\theta^i + \omega_j^i \wedge \theta^j.$$

If we arrange $\theta = (\theta^1, \dots, \theta^n)^T$ as a column vector, then the above formula can be written as

$$\tau = d\theta + \omega \wedge \theta.$$

then τ is called the **torsion 2-form** of ∇ w.r.t. $\{e_i\}$.

Let $T(e_i, e_j) = T_{ij}^k e_k$, where T is the torsion of ∇ . **Prove that**

$$\tau^k(e_i, e_j) = T_{ij}^k.$$

Thus

$$\tau^k = \frac{1}{2} T_{ij}^k \theta^i \wedge \theta^j.$$

Exercise 1.4 (Levi-Civita connection). Let (M, g) be a Riemannian manifold, ∇ be a connection on TM . Show that

1. ∇ is torsion-free if and only if $\tau = 0$, i.e.,

$$d\theta^i + \omega_j^i \wedge \theta^j = 0.$$

2. If $\{e_i\}$ is a local orthonormal frame, then ∇ is compatible with g if and only if ω is skew-symmetric, i.e.,

$$\omega_j^i + \omega_i^j = 0.$$

3. Under a local orthonormal frame, there exists a unique connection 1-form ω satisfying

$$d\theta^i + \omega_j^i \wedge \theta^j = 0, \quad \omega_j^i + \omega_i^j = 0.$$

This is the existence and uniqueness of the Levi-Civita connection.

Exercise 1.5. Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. The following two equations

$$d\theta = \tau - \omega \wedge \theta,$$

$$d\omega = \Omega - \omega \wedge \omega,$$

are called the **structure equations**. Show that τ and Ω satisfy

$$d\tau = \Omega \wedge \theta - \omega \wedge \tau,$$

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega.$$

These two equations are called the **Bianchi identities**. Furthermore, show that

$$d(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

Exercise 1.6. Show that if ∇ is torsion-free, then $d\tau = \Omega \wedge \theta - \omega \wedge \tau$ is equivalent to **the first Bianchi identity**

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0.$$

Exercise 1.7. Show that if ∇ is torsion-free, then $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is equivalent to **the second Bianchi identity**

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Exercise 1.8. For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega_j^k = \Gamma_{ij}^k dx^i$, show that the transformation formula

$$\tilde{\omega} = a^{-1} da + a^{-1} \omega a$$

reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial \tilde{x}^k}{\partial x^l} \frac{\partial x^m}{\partial \tilde{x}^i} \frac{\partial x^n}{\partial \tilde{x}^j} \Gamma_{mn}^l + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j} \frac{\partial \tilde{x}^k}{\partial x^l}.$$

Exercise 1.9. Let ∇ be a connection on TM . It determines a horizontal distribution $\mathcal{H}$ on TM . Then $T(TM) = \mathcal{V} \oplus \mathcal{H}$, where $\mathcal{V}$ is the vertical bundle of TM . Show that ∇ is torsion-free if and only if for any $X, Y \in C^\infty(M, TM)$,

$$[X^h, Y^h] = [X, Y]^h.$$

2 Principal Bundles and Connections

2.1 Principal Bundles

Principal bundles are both a special type of fiber bundle and the most important one. In the following, we will introduce the definition of principal bundles and some basic structures on them.

Definition 2.1 (G -action). A **smooth right action** of a Lie group G on a manifold P is a smooth map

$$\mu : M \times G \rightarrow M,$$

denoted by $\mu(x, g) := x \cdot g$, such that for all $x \in M$ and $g, h \in G$,

$$x \cdot e = x, \quad (x \cdot g) \cdot h = x \cdot (gh).$$

where e is the identity element of G . A manifold M with a smooth right G -action is called a **(right) G -manifold**.

We say the action is **free** if for all $x \in M$, $x \cdot g = x$ implies $g = e$.

Definition 2.2 (G -equivariant map). Let M and N be two G -manifolds. A smooth map $f : M \rightarrow N$ is called **G -equivariant** if for all $x \in M$ and $g \in G$,

$$f(x \cdot g) = f(x) \cdot g.$$

Definition 2.3 (Principal G -bundle). A **principal G -bundle** is a fiber bundle $\pi : P \rightarrow M$ with fiber G together with a free right G -action on P such that for each $x \in M$, the local trivialization

$$\phi_U : \pi^{-1}(U) \rightarrow U \times G,$$

is G -equivariant, where the right G -action on $U \times G$ is given by

$$(x, h) \cdot g = (x, hg), \quad \forall x \in U, g, h \in G.$$

By a **local trivialization** we mean a open neighborhood U of x and a diffeomorphism $\phi_U : \pi^{-1}(U) \rightarrow U \times G$ satisfying the following commutative diagram:

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\phi_U} & U \times G \\ & \searrow \pi & \swarrow \text{pr}_1 \\ & U & \end{array}$$

where $p_1 : U \times G \rightarrow U$ is the projection map.

Example 2.4. Let G be a Lie group and M be a smooth manifold, then the trivial bundle $p_1 : M \times G \rightarrow M$ is a principal G -bundle over M .

Example 2.5. Let G be a Lie group and H be a closed subgroup of G . Then the quotient space G/H is a smooth manifold and the projection map $\pi : G \rightarrow G/H$ is a principal H -bundle over G/H .

Example 2.6. Let $\pi : E \rightarrow M$ be a rank n vector bundle over M . Then the frame bundle $\text{Fr}(E)$ is a principal $\text{GL}(n, \mathbb{R})$ -bundle over M .

Proposition 2.7. Let $\pi : P \rightarrow M$ be a principal G -bundle. Then the right G -action on P is transitive on each fiber.

Proof. Since G acts transitively on $\{x\} \times G$ and the fiber diffeomorphism $\phi_{U,x} : P_x \rightarrow G$ is G -equivariant, thus G acts transitively on P_x . $\square$

Lemma 2.8. For any group G , all right G -equivariant maps are exactly the left translations.

Proof. Let $f : G \rightarrow G$ be a right G -equivariant map, then for all $g \in G$,

$$f(g) = f(e \cdot g) = f(e) \cdot g = l_{f(e)}(g),$$

where $l_{f(e)} : G \rightarrow G$ is the left translation by $f(e)$. $\square$

Definition 2.9 (Cocycle definition). A **principal G -bundle** is a fiber bundle $\pi : P \rightarrow M$ with fiber G together with a local trivialization $\{(U_\alpha, \phi_\alpha)\}$ such that for each $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $g_{\alpha\beta} : U_{\alpha\beta} \rightarrow G$ such that

$$\phi_\alpha \circ \phi_\beta^{-1}(x, h) = (x, g_{\alpha\beta}(x)h), \quad \forall (x, h) \in U_{\alpha\beta} \times G.$$

In addition, the transition functions $\{g_{\alpha\beta}\}$ satisfy the following cocycle condition:

$$\begin{aligned} g_{\alpha\alpha}(x) &= e, \quad \forall x \in U_\alpha, \\ g_{\alpha\beta}(x)g_{\beta\gamma}(x) &= g_{\alpha\gamma}(x), \quad \forall x \in U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset. \end{aligned}$$

Remark. By Lemma 2.8, one can easily check that the two definitions of principal bundles are equivalent. These two definitions characterize G -principal bundles from different perspectives.

2.1.1 Fundamental Vector Fields

Suppose P is a G -manifold, then for each $A \in \mathfrak{g}$, we can define a vector field $\underline{A}$ on P by

$$\underline{A}_p := \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(tA), \quad \forall p \in P.$$

The vector field $\underline{A}$ is called the **fundamental vector field** associated to A .

Fact 2.1. For each $A \in \mathfrak{g}$, $\underline{A}$ is a C^∞ vector field on P .

The fundamental vector field construction gives rise to a map

$$\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P),^1 \quad \sigma(A) := \underline{A}.$$

For $p \in P$, define $j_p : G \rightarrow P$ by $j_p(g) := p \cdot g$, then $j_{p*} := j_{p*e}$ is

$$j_{p*}(A) = \left. \frac{d}{dt} \right|_{t=0} j_p(\exp(tA)) = \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(tA) = \underline{A}_p.$$

This gives another discription of the fundamental vector field: $\underline{A}_p = j_{p*}(A)$. It also implies that the map $\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P)$ is $\mathbb{R}$ -linear. In fact, we have

Fact 2.2. The map $\sigma : \mathfrak{g} \rightarrow \mathfrak{X}(P)$ is a Lie algebra homomorphism, i.e.,

$$[\underline{A}, \underline{B}] = \underline{[A, B]}, \quad \forall A, B \in \mathfrak{g}.$$

Example 2.10. Let G acts on itself by right translation, then the fundamental vector field $\underline{A}$ is exactly the left-invariant vector field generated by A .

For $g \in G$, let $c_g : G \rightarrow G$ be the conjugation map $c_g(h) := ghg^{-1}$. Then the differential of c_g at the identity element e is the adjoint representation $\text{Ad}_g := (c_g)_* : \mathfrak{g} \rightarrow \mathfrak{g}$.

Proposition 2.11. For each $A \in \mathfrak{g}$ and $g \in G$, we have

$$(r_g)_* \underline{A} = \underline{\text{Ad}_{g^{-1}}(A)}.$$

where $r_g : P \rightarrow P$ is the right translation by g , i.e., $r_g(p) := p \cdot g$.

Proof. For each $p \in P$ and $h \in G$, we have

$$(r_g \circ j_p)(h) = r_g(p \cdot h) = (p \cdot h) \cdot g = pg \cdot (g^{-1}hg) = (j_{pg} \circ c_{g^{-1}})(h).$$

By the chain rule,

$$(r_g)_*(\underline{A}_p) = (r_g)_*j_{p*}(A) = j_{pg*}(c_{g^{-1}})_*(A) = j_{pg*}(\text{Ad}_{g^{-1}}(A)) = \underline{\text{Ad}_{g^{-1}}(A)}_{pg}. \quad \square$$

¹ $\mathfrak{X}(P) := C^\infty(P, TP)$ is the space of smooth vector fields on P .

Fact 2.3. Let P be a G -manifold, then the fundamental vector fields $\underline{A}$ vanishes at p if and only if $A \in \text{Lie}(\text{Stab}(p))$. Thus for any $p \in P$, the kernel of $j_{p*} : \mathfrak{g} \rightarrow T_p P$ is exactly $\text{Lie}(\text{Stab}(p))$. In particular, if the G -action on P is free, then j_{p*} is injective for all $p \in P$.

2.1.2 Vertical Subbundle of TP

Let $\pi : P \rightarrow M$ be a principal G -bundle, at each $p \in P$, the differential $\pi_{*p} : T_p P \rightarrow T_{\pi(p)} M$ is surjective. We define $\mathcal{V}_p := \ker \pi_{*p} \subset T_p P$ to be the **vertical tangent space** at p . Then $\mathcal{V} := \bigsqcup_{p \in P} \mathcal{V}_p$ is a subbundle of TP , called the **vertical subbundle** of TP . An element of $\mathcal{V}_p$ is called a **vertical tangent vector** at p .

We have a short exact sequence of vector spaces

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_p P \xrightarrow{\pi_{*p}} T_{\pi(p)} M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_p P - \dim T_{\pi(p)} M = \dim G.$$

Proposition 2.12. *For any $A \in \mathfrak{g}$, the fundamental vector field $\underline{A}$ is vertical at every point $p \in P$.*

Proof. Since $(\pi \circ j_p)(g) = \pi(p \cdot g) = \pi(p)$ is a constant map for all $g \in G$, we have $\pi_{*p}(\underline{A}_p) = \pi_{*p}j_{p*}(A) = (\pi \circ j_p)_*(A) = 0$. $\square$

This proposition implies that $j_{p*}(\mathfrak{g}) \subset \mathcal{V}_p$ for all $p \in P$. In fact, we have the following stronger result.

Proposition 2.13. *For each $p \in P$, the map $j_{p*} : \mathfrak{g} \rightarrow \mathcal{V}_p$ is an isomorphism.*

Proof. Since G acts freely on P , the map j_{p*} is injective. On the other hand, $\dim \mathcal{V}_p = \dim G = \dim \mathfrak{g}$, thus j_{p*} is an isomorphism. $\square$

Corollary 2.14. *The vertical vectors at $p \in P$ are exactly the fundamental vector at p . Moreover, the vertical subbundle $\mathcal{V}$ of TP is isomorphic to the trivial bundle $P \times \mathfrak{g}$.*

Proof. The first statement follows from the previous proposition. For the second statement, let $A_1, \dots, A_l$ be a basis of $\mathfrak{g}$, where $l = \dim G$, then $\underline{A}_1, \dots, \underline{A}_l$ is a global frame of $\mathcal{V}$, thus $\mathcal{V}$ is a trivial bundle. $\square$

The map $\pi_* : TP \rightarrow TM$ induces a bundle map $\tilde{\pi}_* : TP \rightarrow \pi^* TM$ over P . Then we have a short exact sequence of vector bundles over P :

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightarrow{\tilde{\pi}_*} \pi^* TM \longrightarrow 0$$

2.2 Connections on Principal Bundles

2.2.1 Horizontal Distribution

In the category of smooth vector bundles, all short exact sequence of vector bundles splits. Corresponding to the short exact sequence appearing in the previous chapter, there exists a smooth subbundle $\mathcal{H}$ of TP such that $TP = \mathcal{V} \oplus \mathcal{H}$. We call $\mathcal{H}$ a **horizontal distribution** of TP .

In general, there is no canonical choice of $\mathcal{H}$. A choice of $\mathcal{H}$ is the same as a splitting $k : \pi^*TM \rightarrow TP$ such that $\tilde{\pi}_* \circ k = \text{id}_{\pi^*TM}$.

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \begin{array}{c} \xrightarrow{\tilde{\pi}_*} \\ \xleftarrow{k} \end{array} \pi^*TM \longrightarrow 0$$

We say $\mathcal{H}$ is **right-invariant** if for all $g \in G$ and $p \in P$, $r_{g*p}(\mathcal{H}_p) = \mathcal{H}_{pg}$, where $r_g : P \rightarrow P$, $r_g(p) = p \cdot g$.

2.2.2 Ehresmann Connection

Let $\pi : P \rightarrow M$ be a principal G -bundle, $\mathcal{H}$ be a horizontal distribution of TP . For each $p \in P$, we define a projection map $v : T_pP = \mathcal{V}_p \oplus \mathcal{H}_p \rightarrow \mathcal{V}_p$. For $Y_p \in T_pP$, we call $v(Y_p)$ the **vertical component** of Y_p .¹

Let $\iota : \mathcal{V} \hookrightarrow TP$ be the inclusion map, then $v \circ \iota = \text{id}_{\mathcal{V}}$. We actually obtain another splitting of the short exact sequence:

$$0 \longrightarrow \mathcal{V} \begin{array}{c} \xleftarrow{\iota} \\ \xrightarrow{v} \end{array} TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

Recall that for each $p \in P$, the map $j_p : G \rightarrow P$ defined by $j_p(g) = p \cdot g$. We define a $\mathfrak{g}$ -valued 1-form ω on P by

$$\omega_p := j_{p*}^{-1} \circ v : T_pP \xrightarrow{v} \mathcal{V}_p \xrightarrow{j_{p*}^{-1}} \mathfrak{g}.$$

In terms of ω , the vertical component of $Y_p \in T_pP$ is given by

$$v(Y_p) = j_{p*}(\omega_p(Y_p)).$$

In summary, given a projection $v : TP \rightarrow \mathcal{V}$, we induce the following composition of maps:

$$\omega : TP \xrightarrow{v} \mathcal{V} \xrightarrow{j_*^{-1}} P \times \mathfrak{g} \xrightarrow{\text{pr}_2} \mathfrak{g} \rightsquigarrow \omega \in \Omega^1(P, \mathfrak{g}).$$

So the information of the horizontal distribution $\mathcal{H}$ is encoded in ω .

¹Although $\mathcal{V}_p$ is intrinsically defined, the vertical component of a tangent vector depends on the choice of $\mathcal{H}_p$.

Proposition 2.15. *If $\mathcal{H}$ is right-invariant, then the $\mathfrak{g}$ -valued 1-form ω defined above satisfies the following properties:*

1. $\omega_p(\underline{A}_p) = A, \forall p \in P, A \in \mathfrak{g}.$
2. (G -equivalence) $(r_g)^*\omega = (\text{Ad}_{g^{-1}})\omega, \forall g \in G.$
3. ω is smooth.

We say a $\mathfrak{g}$ -valued 1-form ω on P satisfying the above properties is an **Ehresmann connection** or simply a **connection** on P .

Proof. 1. Since $\underline{A}$ is already vertical,

$$\omega_p(\underline{A}_p) = j_{p*}^{-1}(v(\underline{A}_p)) = j_{p*}^{-1}(\underline{A}_p) = A.$$

2. For each $p \in P$ and $Y_p \in T_pP$, we need to show

$$\omega_{pg}((r_g)_*Y_p) = (\text{Ad}_{g^{-1}})\omega_p(Y_p).$$

Since both side are linear in Y_p , we only need to check the above equality for $Y_p \in \mathcal{V}_p$ and $Y_p \in \mathcal{H}_p$ respectively.

If $Y_p \in \mathcal{V}_p$, then $Y_p = \underline{A}_p$ for some $A \in \mathfrak{g}$, thus

$$\omega_{pg}(r_{g*}\underline{A}_p) = \omega_{pg}(\underline{\text{Ad}_{g^{-1}}(A)})_{pg} = \text{Ad}_{g^{-1}}(A) = (\text{Ad}_{g^{-1}})\omega_p(\underline{A}_p).$$

If $Y_p \in \mathcal{H}_p$, by the right-invariance of $\mathcal{H}$, we have $r_{g*}(Y_p) \in \mathcal{H}_{pg}$, thus

$$\omega_{pg}(r_{g*}Y_p) = 0 = (\text{Ad}_{g^{-1}})\omega_p(Y_p).$$

3. Omit. □

Viewing ω as a map $\omega : TP \rightarrow \mathfrak{g}$. Both TP and $\mathfrak{g}$ are G -manifolds, where

$$g \cdot Y_p := (r_g)_*Y_p, \quad g \cdot A := \text{Ad}_{g^{-1}}(A).$$

Then the G -equivalence condition of ω

$$\omega_{pg}((r_g)_*Y_p) = (\text{Ad}_{g^{-1}})\omega_p(Y_p),$$

can be rephrased as: $\omega : TP \rightarrow \mathfrak{g}$ is a G -equivariant map.

2.2.3 Curvature of Connections